Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions	Gaca, 2011	De Feo, 2012	Martínez-Sellés, 2014	Madeira, 2016
0. Study information				
0.1 Author	Gaca	De Feo	Martínez-Sellés	Madeira
0.2 Publication year	2011	2012	2014	2016
0.3 Publication identifier (Title/PMID/DOI)	Outcomes for endocarditis surgery in North America:	The Need for a Specific Risk Prediction System in Nati	Valve surgery in active infective endocarditis: A simple	Assessment of perioperative mortality risk in patients
0.4 Publication journal	J Thorac Cardiovasc Surg	The Scientific World Journal	International Journal of Cardiology	Interactive CardioVascular and Thoracic Surgery
0.5 Model name	STSS score	De Feo score	PALSUSE	modified EuroSCORE-II
1. Participants				
1.1 Were appropriate data sources used?	Probably Yes	Probably Yes	Probably Yes	Probably Yes
1.2 Were all inclusions and exclusions of participants appropriate?	Probably No	Probably Yes	Probably Yes	No Information
Risk of bias introduced by selection of participants	**High RoB**	**High RoB**	**Low RoB**	**Unclear**
Applicability	**Low concern**	**High concern**	**Low concern**	**Unclear**
Relevant information extracted from CHARMS:				
Source of data	*Existing registry*	*Retrospective cohort*	*Existing registry*	*Retrospective cohort*
Recruitment methods	*Selective inclusion*	*Prospective recruitment*	*Consecutive patients*	*No information*
Recruitment dates	*2002 - 2008*	*1980 - 2009*	*2008 - 2010*	*2007 - 2014*
Study setting	*Cardiac surgery centers*	*Cardiac surgery center*	*Cardiac surgery centers*	*Cardiac surgery center*
Inclusion criteria	*All patients with the diagnosis of IE who underwent surgery on the aortic, mitral, and/or tricuspid valves.*	*Infective Endocarditis of native valves*	*Patients with definite or possible IE who underwent surgery.*	*Patients underwent cardiac surgery under extracorporeal circulation for active IE (according to modified Duke criteria).*
Exclusion criteria	*Sites were excluded if data were missing on age, gender, status of surgery, cardiogenic shock , and endocarditis type. And if more than 20% of patients had no complication information reported.*	*Infective Endocarditis of prosthetic valves*	*No information*	*No information*
Rationale of bias and applicability rating:	*Excluding complete sites if data were missing in some variables, likely to have introduced bias but it is less important than to exclude individual participants.*	*The model was developed in a subgroup of patients. These participants represent a selected lower (or higher) risk sample of the original.*		*No information about recruitment methods and exclusion criteria.*
2. Predictors				
2.1 Were predictors defined and assessed in a similar way for all participants?	Yes	Yes	Yes	Yes
2.2 Were predictor assessments made without knowledge of outcome data?	No Information	No Information	No Information	No Information
2.3 Are all predictors available at the time the model is intended to be used?	Yes	Yes	Yes	Yes
Risk of bias introduced by predictors or their assessment	**Low RoB**	**Unclear**	**Low RoB**	**Low RoB**
Applicability	**Low concern**	**Low concern**	**Low concern**	**Low concern**
Relevant information extracted from CHARMS:				
Predictors definition and measurement similar for all participants	*Yes*	*Yes*	*Yes*	*Yes*
Were predictors assessed blinded for outcome?	*No information*	*No information*	*No information*	*No information*
Timing of predictors measurement	*Pre-operative*	*pre-operative*	*Pre-operative*	*Pre-operative*
Rationale of bias and applicability rating:	*Excluded complete sites if data were missing in some variables, likely to have introduced bias but less important than excluding individual participants.*	*There were predictors assessed with subjective interpretation*		
3. Outcome				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions	Gaca, 2011	De Feo, 2012	Martínez-Sellés, 2014	Madeira, 2016
0. Study information				
0.1 Author	Gaca	De Feo	Martínez-Sellés	Madeira
0.2 Publication year	2011	2012	2014	2016
0.3 Publication identifier (Title/PMID/DOI)	Outcomes for endocarditis surgery in North America:	The Need for a Specific Risk Prediction System in Nati	Valve surgery in active infective endocarditis: A simple	Assessment of perioperative mortality risk in patients
0.4 Publication journal	J Thorac Cardiovasc Surg	The Scientific World Journal	International Journal of Cardiology	Interactive CardioVascular and Thoracic Surgery
0.5 Model name	STSS score	De Feo score	PALSUSE	modified EuroSCORE-II
3.1 Was the outcome determined appropriately?	Yes	Yes	Yes	Yes
3.2 Was a pre-specified or standard outcome definition used?	Yes	Yes	Yes	Yes
3.3 Were predictors excluded from the outcome definition?	Yes	Yes	Yes	Yes
3.4 Was the outcome defined and determined in a similar way for all participants?	Yes	Yes	Yes	Yes
3.5 Was the outcome determined without knowledge of predictor information?	No Information	No Information	No Information	No Information
3.6 Was the time interval between predictor assessment and outcome determination appropri	Probably Yes	Probably Yes	Probably Yes	Probably Yes
Risk of bias introduced by the outcome or its determination	**Low RoB**	**Low RoB**	**Low RoB**	**Low RoB**
Applicability	**Low concern**	**Low concern**	**Low concern**	**Low concern**
Relevant information extracted from CHARMS:				
Outcome definition	*Death occurring before discharge or within 30 days of surgery.*	*30-day death or in hospital death*	*Operative mortality ocurring during the admission*	*at 30 days or during the index hospitalization*
Same outcome definition for all participants	*Yes*	*Yes*	*Yes*	*Yes*
Was the outcome assessed without knowledge of the predictors	*Unclear*	*Unclear*	*Unclear*	*Unclear*
Were candidate predictors part of outcome?	*No*	*No*	*No*	*No information*
Time of outcome ocurrence	*30 days or lenght of hospital stay*	*30 days or lenght of hospital stay*	*Lenght of hospital stay*	*30 days or lenght of hospital stay*
Rationale of bias and applicability rating:				
4. Analysis				
4.1 Were there a reasonable number of participants with the outcome?	Yes	No	No Information	No
4.2 Were continuous and categorical predictors handled appropriately?	Probably Yes	No	Probably No	Probably No
4.3 Were all enrolled participants included in the analysis?	Probably No	Probably No	Probably Yes	Probably Yes
4.4 Were participants with missing data handled appropriately?	Probably No	No Information	No Information	No Information
4.5 Was selection of predictors based on univariable analysis avoided?	No	No	No	No
4.6 Were complexities in the data accounted for appropriately?	Probably Yes	Yes	Yes	Yes
4.7 Were relevant model performance measures evaluated appropriately?	Probably No	No	No	No
4.8 Were model overfitting and optimism in model performance accounted for?	No	No	No	No
4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis?	Yes	Probably No	Probably No	Yes
Risk of bias introduced by the analysis	**High RoB**	**High RoB**	**High RoB**	**High RoB**
Relevant information extracted from CHARMS:				
Outcome frecuency	*1117 from 13617 (8.2%)*	*40 from 440 (9.1%)*	*106 from 437 (24.3%)*	*21 from 128 (16.4%)*
Event per variable (EPV) or per parameter (EPP)	*29.4*	*2.1*	*Unknown*	*1.4*
Handling of continuous predictors	*No information*	*No information*	*Dichotomization*	*No information*
Number of participants with any missing value	*98*	*22*	*No information*	*No information*

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions	Gaca, 2011	De Feo, 2012	Martínez-Sellés, 2014	Madeira, 2016
0. Study information				
0.1 Author	Gaca	De Feo	Martínez-Sellés	Madeira
0.2 Publication year	2011	2012	2014	2016
0.3 Publication identifier (Title/PMID/DOI)	Outcomes for endocarditis surgery in North America:	The Need for a Specific Risk Prediction System in Nati	Valve surgery in active infective endocarditis: A simple	Assessment of perioperative mortality risk in patients
0.4 Publication journal	J Thorac Cardiovasc Surg	The Scientific World Journal	International Journal of Cardiology	Interactive CardioVascular and Thoracic Surgery
0.5 Model name	STSS score	De Feo score	PALSUSE	modified EuroSCORE-II
Handling of missing data	*No information*	*No information*	*No information*	*No information*
Method for selection of candidate predictors	*Based on univariable associations*	*Based on univariable associations*	*Based on univariable associations*	*Based on univariable associations*
Validation method	*Internal:* *No* *External:* *1*	*Internal:* *No* *External:* *1*	*Internal:* *No* *External:* *1*	*Internal:* *No* *External:* *1*
Performance measures	*Calibration:* *Calibration plot* *Discrimination:* *C-Statistic* *Overall:* *Not evaluated*	*Calibration:* *HL test* *Discrimination:* *C-Statistic / AUC graph* *Overall:* *Not evaluated*	*Calibration:* *HL test* *Discrimination:* *C-Statistic / AUC graph* *Overall:* *Not evaluated*	*Calibration:* *Calibration plot / Slope / CITL / HL test* *Discrimination:* *C-Statistic / AUC graph* *Overall:* *Brier score*
Shrinkage of predictor weights or regression coefficients	*No*	*No*	*No*	*No*
Rationale of bias and applicability rating:	*Large EPV (aprox. 30), but predictors selected based on univariable analysis, random split sample (D:70% and V: 30%) and no inform how missing data were handled.*	*Very small EPV, continuous predictors no handled apropiately, probably using complete data and only apparent validation available.*	*EPP not available, no informed about missing data, continuous predictors dichotomized and apparent performance.*	*Very small EPP and apparent validation.*

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions	Gatti (a), 2017	Gatti (b), 2017	Di Mauro, 2017	Gatti (c), 2017
0. Study information				
0.1 Author	Gatti (a)	Gatti (b)	Di Mauro	Gatti (c)
0.2 Publication year	2017	2017	2017	2017
0.3 Publication identifier (Title/PMID/DOI)	Simple Scoring System to Predict In-Hospital Mortality	Simple Scoring System to Predict In-Hospital Mortality	A predictive model for early mortality after surgical tr	A risk factor analysis for in-hospital mortality after sur
0.4 Publication journal	Journal of American Heart Association	Journal of American Heart Association	Int J Cardiol	Infection
0.5 Model name	AEPEI score	Alternate AEPEI score	EndoSCORE	ANCLA score
1. Participants				
1.1 Were appropriate data sources used?	Probably Yes	Probably Yes	Probably Yes	Probably Yes
1.2 Were all inclusions and exclusions of participants appropriate?	Probably Yes	Probably Yes	No Information	Probably Yes
Risk of bias introduced by selection of participants	**Low RoB**	**Low RoB**	**Unclear**	**Low RoB**
Applicability	**Low concern**	**Low concern**	**Unclear**	**Low concern**
Relevant information extracted from CHARMS:				
Source of data	*Other (specify)*	*Other (specify)*	*Retrospective cohort*	*Retrospective cohort*
Recruitment methods	*Consecutive patients*	*Consecutive patients*	*No information*	*Consecutive patients*
Recruitment dates	*2000 - 2015 (Italy) 2008 (France)*	*2000 - 2015 (Italy) 2008 (France)*	*2000 - 2015*	*1999 - 2015*
Study setting	*Cardiac surgery centers*	*Cardiac surgery centers*	*Cardiac surgery centers*	*Cardiac surgery center*
Inclusion criteria	*Patients who underwent surgery for IE defined by the modified Duke criteria*	*Patients who underwent surgery for IE defined by the modified Duke criteria*	*Patients with native valve endocarditis (NVE) or prosthesis valve endocarditis (PVE) operated.*	*Patients who underwent surgery for IE defined by the modified Duke criteria.*
Exclusion criteria	*No information*	*No information*	*No information*	*No information*
Rationale of bias and applicability rating:	*The model was developed using only data from 2008 in France, because the data collection was particularly exhaustive, comprehensive, and accurate, we did not consider it could introduce bias.*	*The model was developed using only data from 2008 in France, because the data collection was particularly exhaustive, comprehensive, and accurate, we did not consider it could introduce bias.*	*No information about recruitment methods and exclusion criteria.*	
2. Predictors				
2.1 Were predictors defined and assessed in a similar way for all participants?	Probably Yes	Probably Yes	Yes	Yes
2.2 Were predictor assessments made without knowledge of outcome data?	No Information	No Information	No Information	No Information
2.3 Are all predictors available at the time the model is intended to be used?	Probably No	Yes	Yes	Yes
Risk of bias introduced by predictors or their assessment	**Low RoB**	**Low RoB**	**Low RoB**	**Low RoB**
Applicability	**Unclear**	**Low concern**	**Low concern**	**Low concern**
Relevant information extracted from CHARMS:				
Predictors definition and measurement similar for all participants	*Unclear*	*Unclear*	*Yes*	*Yes*
Were predictors assessed blinded for outcome?	*No information*	*No information*	*No information*	*No information*
Timing of predictors measurement	*Pre-operative*	*Pre-operative*	*Pre-operative*	*Pre-operative*
Rationale of bias and applicability rating:	*Systolic pulmonary artery pressure predictor could be hard to recovery.*			
3. Outcome				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions	Gatti (a), 2017	Gatti (b), 2017	Di Mauro, 2017	Gatti (c), 2017
0. Study information				
0.1 Author	Gatti (a)	Gatti (b)	Di Mauro	Gatti (c)
0.2 Publication year	2017	2017	2017	2017
0.3 Publication identifier (Title/PMID/DOI)	Simple Scoring System to Predict In-Hospital Mortality	Simple Scoring System to Predict In-Hospital Mortality	A predictive model for early mortality after surgical tr	A risk factor analysis for in-hospital mortality after sur
0.4 Publication journal	Journal of American Heart Association	Journal of American Heart Association	Int J Cardiol	Infection
0.5 Model name	AEPEI score	Alternate AEPEI score	EndoSCORE	ANCLA score
3.1 Was the outcome determined appropriately?	Yes	Yes	Yes	Yes
3.2 Was a pre-specified or standard outcome definition used?	Yes	Yes	Yes	Yes
3.3 Were predictors excluded from the outcome definition?	Yes	Yes	Yes	Yes
3.4 Was the outcome defined and determined in a similar way for all participants?	Yes	Yes	Yes	Yes
3.5 Was the outcome determined without knowledge of predictor information?	No Information	No Information	No Information	No Information
3.6 Was the time interval between predictor assessment and outcome determination appropri	Probably Yes	Probably Yes	Probably Yes	Probably Yes
Risk of bias introduced by the outcome or its determination	**Low RoB**	**Low RoB**	**Low RoB**	**Low RoB**
Applicability	**Low concern**	**Low concern**	**Low concern**	**Low concern**
Relevant information extracted from CHARMS:				
Outcome definition	*In-hospital death*	*In-hospital death*	*Death by 30 days after surgery due to any cause*	*In-hospital postsurgery mortality*
Same outcome definition for all participants	*Yes*	*Yes*	*Yes*	*Yes*
Was the outcome assessed without knowledge of the predictors	*Unclear*	*Unclear*	*Unclear*	*Unclear*
Were candidate predictors part of outcome?	*No*	*No*	*No information*	*No*
Time of outcome ocurrence	*Lenght of hospital stay*	*Lenght of hospital stay*	*30 days*	*Lenght of hospital stay*
Rationale of bias and applicability rating:				
4. Analysis				
4.1 Were there a reasonable number of participants with the outcome?	No	No	Probably No	No
4.2 Were continuous and categorical predictors handled appropriately?	No	No	Probably Yes	No
4.3 Were all enrolled participants included in the analysis?	No Information	Probably No	Probably Yes	Probably No
4.4 Were participants with missing data handled appropriately?	No Information	No Information	No Information	No Information
4.5 Was selection of predictors based on univariable analysis avoided?	No	No	No	No
4.6 Were complexities in the data accounted for appropriately?	Yes	Yes	Yes	Yes
4.7 Were relevant model performance measures evaluated appropriately?	Yes	Probably No	Yes	No
4.8 Were model overfitting and optimism in model performance accounted for?	No	No	Yes	No
4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis?	Yes	Yes	Yes	Probably No
Risk of bias introduced by the analysis	**High RoB**	**High RoB**	**Unclear**	**High RoB**
Relevant information extracted from CHARMS:				
Outcome frecuency	*56 from 361 (15.5%)*	*56 from 361 (15.5%)*	*298 from 2715 (11.0%)*	*28 from 138 (20.3%)*
Event per variable (EPV) or per parameter (EPP)	*1.0*	*1.0*	*9.3*	*0.5*
Handling of continuous predictors	*No information*	*No information*	*Restricted cubic spline function or categorization*	*No information*
Number of participants with any missing value	*No information*	*No information*	*No information*	*45*

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions	Gatti (a), 2017	Gatti (b), 2017	Di Mauro, 2017	Gatti (c), 2017
0. Study information				
0.1 Author	Gatti (a)	Gatti (b)	Di Mauro	Gatti (c)
0.2 Publication year	2017	2017	2017	2017
0.3 Publication identifier (Title/PMID/DOI)	Simple Scoring System to Predict In-Hospital Mortality	Simple Scoring System to Predict In-Hospital Mortality	A predictive model for early mortality after surgical tr	A risk factor analysis for in-hospital mortality after sur
0.4 Publication journal	Journal of American Heart Association	Journal of American Heart Association	Int J Cardiol	Infection
0.5 Model name	AEPEI score	Alternate AEPEI score	EndoSCORE	ANCLA score
Handling of missing data	*No information*	*No information*	*No information*	*No information*
Method for selection of candidate predictors	*Based on univariable associations*	*Based on univariable associations*	*Based on univariable associations*	*Based on univariable associations*
Validation method	*Internal:* *No* *External:* *1*	*Internal:* *No* *External:* *1*	*Internal:* *No* *External:* *1*	*Internal:* *No* *External:* *1*
Performance measures	*Calibration:* *HL test* *Discrimination:* *C-Statistic / AUC graph* *Overall:* *Not evaluated*	*Calibration:* *HL test* *Discrimination:* *C-Statistic / AUC graph* *Overall:* *Not evaluated*	*Calibration:* *Comparison of actual CITL and slope with the ideal values* *Discrimination:* *C-Statistic / AUC graph* *Overall:* *Brier score*	*Calibration:* *HL test* *Discrimination:* *C-Statistic / AUC graph* *Overall:* *Not evaluated*
Shrinkage of predictor weights or regression coefficients	*No*	*No*	*No*	*No*
Rationale of bias and applicability rating:	*Very small EPP, predictors selected based on univariable analysis and continuous predictors categorized based on the best discriminative performance.*	*Very small EPP, predictors selected based on univariable analysis and continuous predictors categorized based on the best discriminative performance.*	*Predictors were selected based on univariable analysis (p<0.2). Although EPP was sufficiently large and model performance was optimism adjusted, unfortunately calibration measures were tested but not reported, thus we rated it as unclear RoB.*	*Very small EPP, predictors selected based on univariable analysis and continuous predictors categorized based on the best discriminative performance.*

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions	Olmos, 2017	Fernández-Hidalgo (a), 2018	Fernández-Hidalgo (b), 2018	
0. Study information				
0.1 Author	Olmos	Fernández-Hidalgo (a)	Fernández-Hidalgo (b)	
0.2 Publication year	2017	2018	2018	
0.3 Publication identifier (Title/PMID/DOI)	Risk score for cardiac surgery in active left-sided infec	A pragmatic approach for mortality prediction after su	A pragmatic approach for mortality prediction after su	
0.4 Publication journal	Heart	Clinical Microbiology and Infection	Clinical Microbiology and Infection	
0.5 Model name	RISK-E score	Specific EuroSCORE-I	Specific EuroSCORE-II	
1. Participants				
1.1 Were appropriate data sources used?	Probably Yes	Probably Yes	Probably Yes	
1.2 Were all inclusions and exclusions of participants appropriate?	Probably Yes	Probably Yes	Probably Yes	
Risk of bias introduced by selection of participants	**Low RoB**	**Low RoB**	**Low RoB**	
Applicability	**Low concern**	**Low concern**	**Low concern**	
Relevant information extracted from CHARMS:				
Source of data	*Retrospective cohort*	*Retrospective cohort*	*Retrospective cohort*	
Recruitment methods	*Consecutive patients prospectively recruited on an*	*Consecutive patients*	*Consecutive patients*	
Recruitment dates	*1996 - 2014*	*2000 - 2011*	*2000 - 2011*	
Study setting	*Cardiac surgery centers*	*Cardiac surgery centers*	*Cardiac surgery centers*	
Inclusion criteria	*Patients with definite, left-sided IE in the active phase of the disease who underwent surgery.*	*Patients who requiring cardiac surgery during the active phase of infection (while on antimicrobial therapy).*	*Patients who requiring cardiac surgery during the active phase of infection (while on antimicrobial therapy).*	
Exclusion criteria	*Patients with right-sided IE, left-sided IE without surgical indications and those with a surgical indication that were not surgical candidates.*	*IE cases exclusively involving pacemaker leads were not considered. Only the first episode of IE recorded for an individual patient was used.*	*IE cases exclusively involving pacemaker leads were not considered. Only the first episode of IE recorded for an individual patient was used.*	
Rationale of bias and applicability rating:				
2. Predictors				
2.1 Were predictors defined and assessed in a similar way for all participants?	Yes	No	No	
2.2 Were predictor assessments made without knowledge of outcome data?	No Information	No Information	No Information	
2.3 Are all predictors available at the time the model is intended to be used?	Yes	Yes	Yes	
Risk of bias introduced by predictors or their assessment	**Low RoB**	**Low RoB**	**Low RoB**	
Applicability	**Low concern**	**Unclear**	**Low concern**	
Relevant information extracted from CHARMS:				
Predictors definition and measurement similar for all participants	*Yes*	*No*	*No*	
Were predictors assessed blinded for outcome?	*No information*	*No information*	*No information*	
Timing of predictors measurement	*Pre-operative*	*Pre-operative*	*Pre-operative*	
Rationale of bias and applicability rating:		*Databases were not homogeneous, but authors did an effort to homogenize it, we did not penalized the RoB. Systolic pulmonary artery pressure predictor could be hard to recovery.*	*Databases were not homogeneous, but authors did an effort to homogenize it, we did not penalized the RoB.*	
3. Outcome				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions	Olmos, 2017	Fernández-Hidalgo (a), 2018	Fernández-Hidalgo (b), 2018	
0. Study information				
0.1 Author	Olmos	Fernández-Hidalgo (a)	Fernández-Hidalgo (b)	
0.2 Publication year	2017	2018	2018	
0.3 Publication identifier (Title/PMID/DOI)	Risk score for cardiac surgery in active left-sided infec	A pragmatic approach for mortality prediction after su	A pragmatic approach for mortality prediction after su	
0.4 Publication journal	Heart	Clinical Microbiology and Infection	Clinical Microbiology and Infection	
0.5 Model name	RISK-E score	Specific EuroSCORE-I	Specific EuroSCORE-II	
3.1 Was the outcome determined appropriately?	Yes	Yes	Yes	
3.2 Was a pre-specified or standard outcome definition used?	Yes	Yes	Yes	
3.3 Were predictors excluded from the outcome definition?	Yes	Yes	Yes	
3.4 Was the outcome defined and determined in a similar way for all participants?	Yes	Yes	Yes	
3.5 Was the outcome determined without knowledge of predictor information?	No Information	No Information	No Information	
3.6 Was the time interval between predictor assessment and outcome determination appropri	Probably Yes	Probably Yes	Probably Yes	
Risk of bias introduced by the outcome or its determination	**Low RoB**	**Low RoB**	**Low RoB**	
Applicability	**Low concern**	**Low concern**	**Low concern**	
Relevant information extracted from CHARMS:				
Outcome definition	*Death ocurring after surgery and before discharge, regardless of its cause.*	*Death occurring within the first 30 days after cardiac surgery.*	*Death occurring within the first 30 days after cardiac surgery.*	
Same outcome definition for all participants	*Yes*	*Yes*	*Yes*	
Was the outcome assessed without knowledge of the predictors	*Unclear*	*Unclear*	*Unclear*	
Were candidate predictors part of outcome?	*No*	*No*	*No*	
Time of outcome ocurrence	*Lenght of hospital stay*	*30 days*	*30 days*	
Rationale of bias and applicability rating:				
4. Analysis				
4.1 Were there a reasonable number of participants with the outcome?	No	Probably No	Probably No	
4.2 Were continuous and categorical predictors handled appropriately?	No	Probably Yes	Probably Yes	
4.3 Were all enrolled participants included in the analysis?	No	Probably No	Probably No	
4.4 Were participants with missing data handled appropriately?	No Information	No Information	No Information	
4.5 Was selection of predictors based on univariable analysis avoided?	No	Yes	Yes	
4.6 Were complexities in the data accounted for appropriately?	Yes	Yes	Yes	
4.7 Were relevant model performance measures evaluated appropriately?	Yes	Yes	Yes	
4.8 Were model overfitting and optimism in model performance accounted for?	No	Yes	Yes	
4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis?	Yes	Yes	Yes	
Risk of bias introduced by the analysis	**High RoB**	**Low RoB**	**Low RoB**	
Relevant information extracted from CHARMS:				
Outcome frecuency	*124 from 424 (29.2%)*	*208 from 779 (26.7%)*	*208 from 779 (26.7%)*	
Event per variable (EPV) or per parameter (EPP)	*3.4*	*8.0*	*7.7*	
Handling of continuous predictors	*No information*	*No information*	*No information*	
Number of participants with any missing value	*No information*	*4*	*4*	

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions	Olmos, 2017	Fernández-Hidalgo (a), 2018	Fernández-Hidalgo (b), 2018	
0. Study information				
0.1 Author	Olmos	Fernández-Hidalgo (a)	Fernández-Hidalgo (b)	
0.2 Publication year	2017	2018	2018	
0.3 Publication identifier (Title/PMID/DOI)	Risk score for cardiac surgery in active left-sided infec	A pragmatic approach for mortality prediction after su	A pragmatic approach for mortality prediction after su	
0.4 Publication journal	Heart	Clinical Microbiology and Infection	Clinical Microbiology and Infection	
0.5 Model name	RISK-E score	Specific EuroSCORE-I	Specific EuroSCORE-II	
Handling of missing data	*No information*	*Complete-case analysis*	*Complete-case analysis*	
Method for selection of candidate predictors	*Based on univariable associations and clinical relevance*	*Based on prior knowledge*	*Based on prior knowledge*	
Validation method	*Internal:* *No* *External:* *1*	*Internal:* *No* *External:* *1*	*Internal:* *No* *External:* *1*	*External:*
Performance measures	*Calibration:* *Calibration plot / HL test* *Discrimination:* *C-Statistic / AUC graph* *Overall:* *Not evaluated*	*Calibration:* *Calibration plot / Slope / CITL* *Discrimination:* *C-Statistic / AUC graph* *Overall:* *Brier score*	*Calibration:* *Calibration plot / Slope / CITL* *Discrimination:* *C-Statistic / AUC graph* *Overall:* *Brier score*	
Shrinkage of predictor weights or regression coefficients	*No*	*No*	*No*	
Rationale of bias and applicability rating:	*Very small EPP and random splitting validation (D:70% and V:30%) and did not inform neither missing data nor continuous predictors were handled.*	*EPP was slightly lower than required but were not univariable selection and model performance was optimism adjusted by bootstrap validation. The complete data analysis were not worrying because only 4 (0.5%) patients were excluded. We did not penalized RoB.*	*EPP was slightly lower than required but were not univariable selection and model performance was optimism adjusted by bootstrap validation. The complete data analysis were not worrying because only 4 (0.5%) patients were excluded. We did not penalized RoB.*	

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions				
0. Study information				
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name				
1. Participants				
1.1 Were appropriate data sources used?				
1.2 Were all inclusions and exclusions of participants appropriate?				
Risk of bias introduced by selection of participants				
Applicability				
Relevant information extracted from CHARMS: *Source of data* *Recruitment methods* *Recruitment dates* *Study setting* *Inclusion criteria* *Exclusion criteria*				
Rationale of bias and applicability rating:				
2. Predictors				
2.1 Were predictors defined and assessed in a similar way for all participants?				
2.2 Were predictor assessments made without knowledge of outcome data?				
2.3 Are all predictors available at the time the model is intended to be used?				
Risk of bias introduced by predictors or their assessment				
Applicability				
Relevant information extracted from CHARMS: *Predictors definition and measurement similar for all participants* *Were predictors assessed blinded for outcome?* *Timing of predictors measurement*				
Rationale of bias and applicability rating:				
3. Outcome				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions				
0. Study information				
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name				
3.1 Was the outcome determined appropriately?				
3.2 Was a pre-specified or standard outcome definition used?				
3.3 Were predictors excluded from the outcome definition?				
3.4 Was the outcome defined and determined in a similar way for all participants?				
3.5 Was the outcome determined without knowledge of predictor information?				
3.6 Was the time interval between predictor assessment and outcome determination appropri				
Risk of bias introduced by the outcome or its determination				
Applicability				
Relevant information extracted from CHARMS: *Outcome definition* *Same outcome definition for all participants* *Was the outcome assessed without knowledge of the predictors* *Were candidate predictors part of outcome?* *Time of outcome ocurrence*				
Rationale of bias and applicability rating:				
4. Analysis				
4.1 Were there a reasonable number of participants with the outcome?				
4.2 Were continuous and categorical predictors handled appropriately?				
4.3 Were all enrolled participants included in the analysis?				
4.4 Were participants with missing data handled appropriately?				
4.5 Was selection of predictors based on univariable analysis avoided?				
4.6 Were complexities in the data accounted for appropriately?				
4.7 Were relevant model performance measures evaluated appropriately?				
4.8 Were model overfitting and optimism in model performance accounted for?				
4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis?				
Risk of bias introduced by the analysis				
Relevant information extracted from CHARMS: *Outcome frecuency* *Event per variable (EPV) or per parameter (EPP)* *Handling of continuous predictors* *Number of participants with any missing value*				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions				
0. Study information				
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name *Handling of missing data* *Method for selection of candidate predictors* *Validation method* *Performance measures* *Shrinkage of predictor weights or regression coefficients*	*External:*	*External:*	*External:*	*External:*
Rationale of bias and applicability rating:				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions				
0. Study information				
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name				
1. Participants				
1.1 Were appropriate data sources used?				
1.2 Were all inclusions and exclusions of participants appropriate?				
Risk of bias introduced by selection of participants				
Applicability				
Relevant information extracted from CHARMS: *Source of data* *Recruitment methods* *Recruitment dates* *Study setting* *Inclusion criteria* *Exclusion criteria*				
Rationale of bias and applicability rating:				
2. Predictors				
2.1 Were predictors defined and assessed in a similar way for all participants?				
2.2 Were predictor assessments made without knowledge of outcome data?				
2.3 Are all predictors available at the time the model is intended to be used?				
Risk of bias introduced by predictors or their assessment				
Applicability				
Relevant information extracted from CHARMS: *Predictors definition and measurement similar for all participants* *Were predictors assessed blinded for outcome?* *Timing of predictors measurement*				
Rationale of bias and applicability rating:				
3. Outcome				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions				
0. Study information				
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name				
3.1 Was the outcome determined appropriately?				
3.2 Was a pre-specified or standard outcome definition used?				
3.3 Were predictors excluded from the outcome definition?				
3.4 Was the outcome defined and determined in a similar way for all participants?				
3.5 Was the outcome determined without knowledge of predictor information?				
3.6 Was the time interval between predictor assessment and outcome determination appropri				
Risk of bias introduced by the outcome or its determination				
Applicability				
Relevant information extracted from CHARMS: *Outcome definition* *Same outcome definition for all participants* *Was the outcome assessed without knowledge of the predictors* *Were candidate predictors part of outcome?* *Time of outcome ocurrence*				
Rationale of bias and applicability rating:				
4. Analysis				
4.1 Were there a reasonable number of participants with the outcome?				
4.2 Were continuous and categorical predictors handled appropriately?				
4.3 Were all enrolled participants included in the analysis?				
4.4 Were participants with missing data handled appropriately?				
4.5 Was selection of predictors based on univariable analysis avoided?				
4.6 Were complexities in the data accounted for appropriately?				
4.7 Were relevant model performance measures evaluated appropriately?				
4.8 Were model overfitting and optimism in model performance accounted for?				
4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis?				
Risk of bias introduced by the analysis				
Relevant information extracted from CHARMS: *Outcome frecuency* *Event per variable (EPV) or per parameter (EPP)* *Handling of continuous predictors* *Number of participants with any missing value*				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions				
0. Study information				
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name *Handling of missing data* *Method for selection of candidate predictors* *Validation method* *Performance measures* *Shrinkage of predictor weights or regression coefficients*	*External:*	*External:*	*External:*	*External:*
Rationale of bias and applicability rating:				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions				
0. Study information				
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name				
1. Participants				
1.1 Were appropriate data sources used?				
1.2 Were all inclusions and exclusions of participants appropriate?				
Risk of bias introduced by selection of participants				
Applicability				
Relevant information extracted from CHARMS: *Source of data* *Recruitment methods* *Recruitment dates* *Study setting* *Inclusion criteria* *Exclusion criteria*				
Rationale of bias and applicability rating:				
2. Predictors				
2.1 Were predictors defined and assessed in a similar way for all participants?				
2.2 Were predictor assessments made without knowledge of outcome data?				
2.3 Are all predictors available at the time the model is intended to be used?				
Risk of bias introduced by predictors or their assessment				
Applicability				
Relevant information extracted from CHARMS: *Predictors definition and measurement similar for all participants* *Were predictors assessed blinded for outcome?* *Timing of predictors measurement*				
Rationale of bias and applicability rating:				
3. Outcome				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions				
0. Study information				
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name				
3.1 Was the outcome determined appropriately?				
3.2 Was a pre-specified or standard outcome definition used?				
3.3 Were predictors excluded from the outcome definition?				
3.4 Was the outcome defined and determined in a similar way for all participants?				
3.5 Was the outcome determined without knowledge of predictor information?				
3.6 Was the time interval between predictor assessment and outcome determination appropri				
Risk of bias introduced by the outcome or its determination				
Applicability				
Relevant information extracted from CHARMS: *Outcome definition* *Same outcome definition for all participants* *Was the outcome assessed without knowledge of the predictors* *Were candidate predictors part of outcome?* *Time of outcome ocurrence*				
Rationale of bias and applicability rating:				
4. Analysis				
4.1 Were there a reasonable number of participants with the outcome?				
4.2 Were continuous and categorical predictors handled appropriately?				
4.3 Were all enrolled participants included in the analysis?				
4.4 Were participants with missing data handled appropriately?				
4.5 Was selection of predictors based on univariable analysis avoided?				
4.6 Were complexities in the data accounted for appropriately?				
4.7 Were relevant model performance measures evaluated appropriately?				
4.8 Were model overfitting and optimism in model performance accounted for?				
4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis?				
Risk of bias introduced by the analysis				
Relevant information extracted from CHARMS: *Outcome frecuency* *Event per variable (EPV) or per parameter (EPP)* *Handling of continuous predictors* *Number of participants with any missing value*				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions				
0. Study information				
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name *Handling of missing data* *Method for selection of candidate predictors* *Validation method* *Performance measures* *Shrinkage of predictor weights or regression coefficients*	*External:*	*External:*	*External:*	*External:*
Rationale of bias and applicability rating:				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions				
0. Study information				
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name				
1. Participants				
1.1 Were appropriate data sources used?				
1.2 Were all inclusions and exclusions of participants appropriate?				
Risk of bias introduced by selection of participants				
Applicability				
Relevant information extracted from CHARMS: *Source of data* *Recruitment methods* *Recruitment dates* *Study setting* *Inclusion criteria* *Exclusion criteria*				
Rationale of bias and applicability rating:				
2. Predictors				
2.1 Were predictors defined and assessed in a similar way for all participants?				
2.2 Were predictor assessments made without knowledge of outcome data?				
2.3 Are all predictors available at the time the model is intended to be used?				
Risk of bias introduced by predictors or their assessment				
Applicability				
Relevant information extracted from CHARMS: *Predictors definition and measurement similar for all participants* *Were predictors assessed blinded for outcome?* *Timing of predictors measurement*				
Rationale of bias and applicability rating:				
3. Outcome				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions				
0. Study information				
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name				
3.1 Was the outcome determined appropriately?				
3.2 Was a pre-specified or standard outcome definition used?				
3.3 Were predictors excluded from the outcome definition?				
3.4 Was the outcome defined and determined in a similar way for all participants?				
3.5 Was the outcome determined without knowledge of predictor information?				
3.6 Was the time interval between predictor assessment and outcome determination appropri				
Risk of bias introduced by the outcome or its determination				
Applicability				
Relevant information extracted from CHARMS: *Outcome definition* *Same outcome definition for all participants* *Was the outcome assessed without knowledge of the predictors* *Were candidate predictors part of outcome?* *Time of outcome ocurrence*				
Rationale of bias and applicability rating:				
4. Analysis				
4.1 Were there a reasonable number of participants with the outcome?				
4.2 Were continuous and categorical predictors handled appropriately?				
4.3 Were all enrolled participants included in the analysis?				
4.4 Were participants with missing data handled appropriately?				
4.5 Was selection of predictors based on univariable analysis avoided?				
4.6 Were complexities in the data accounted for appropriately?				
4.7 Were relevant model performance measures evaluated appropriately?				
4.8 Were model overfitting and optimism in model performance accounted for?				
4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis?				
Risk of bias introduced by the analysis				
Relevant information extracted from CHARMS: *Outcome frecuency* *Event per variable (EPV) or per parameter (EPP)* *Handling of continuous predictors* *Number of participants with any missing value*				

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions				
0. Study information				
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name *Handling of missing data* *Method for selection of candidate predictors* *Validation method* *Performance measures* *Shrinkage of predictor weights or regression coefficients*	*External:*	*External:*	*External:*	*External:*
Rationale of bias and applicability rating:				

Home | Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions		
0. Study information		
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name		
1. Participants		
1.1 Were appropriate data sources used?		
1.2 Were all inclusions and exclusions of participants appropriate?		
Risk of bias introduced by selection of participants		
Applicability		
Relevant information extracted from CHARMS: *Source of data* *Recruitment methods* *Recruitment dates* *Study setting* *Inclusion criteria* *Exclusion criteria*		
Rationale of bias and applicability rating:		
2. Predictors		
2.1 Were predictors defined and assessed in a similar way for all participants?		
2.2 Were predictor assessments made without knowledge of outcome data?		
2.3 Are all predictors available at the time the model is intended to be used?		
Risk of bias introduced by predictors or their assessment		
Applicability		
Relevant information extracted from CHARMS: *Predictors definition and measurement similar for all participants* *Were predictors assessed blinded for outcome?* *Timing of predictors measurement*		
Rationale of bias and applicability rating:		
3. Outcome		

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions		
0. Study information		
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name		
3.1 Was the outcome determined appropriately?		
3.2 Was a pre-specified or standard outcome definition used?		
3.3 Were predictors excluded from the outcome definition?		
3.4 Was the outcome defined and determined in a similar way for all participants?		
3.5 Was the outcome determined without knowledge of predictor information?		
3.6 Was the time interval between predictor assessment and outcome determination appropri		
Risk of bias introduced by the outcome or its determination		
Applicability		
Relevant information extracted from CHARMS: *Outcome definition* *Same outcome definition for all participants* *Was the outcome assessed without knowledge of the predictors* *Were candidate predictors part of outcome?* *Time of outcome ocurrence*		
Rationale of bias and applicability rating:		
4. Analysis		
4.1 Were there a reasonable number of participants with the outcome?		
4.2 Were continuous and categorical predictors handled appropriately?		
4.3 Were all enrolled participants included in the analysis?		
4.4 Were participants with missing data handled appropriately?		
4.5 Was selection of predictors based on univariable analysis avoided?		
4.6 Were complexities in the data accounted for appropriately?		
4.7 Were relevant model performance measures evaluated appropriately?		
4.8 Were model overfitting and optimism in model performance accounted for?		
4.9 Do predictors and their assigned weights in the final model correspond to the results from multivariable analysis?		
Risk of bias introduced by the analysis		
Relevant information extracted from CHARMS: *Outcome frecuency* *Event per variable (EPV) or per parameter (EPP)* *Handling of continuous predictors* *Number of participants with any missing value*		

Home Summary

PROBAST checklist

PROBAST. Prediction model study Risk Of Bias Assessment Tool

Domain/ Key questions		
0. Study information		
0.1 Author 0.2 Publication year 0.3 Publication identifier (Title/PMID/DOI) 0.4 Publication journal 0.5 Model name *Handling of missing data* *Method for selection of candidate predictors* *Validation method* *Performance measures* *Shrinkage of predictor weights or regression coefficients*	*External:*	*External:*
Rationale of bias and applicability rating:		